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Short communication

Cyathane diterpenes from Chinese mushroom *Sarcodon scabrosus* and their neurite outgrowth-promoting activity

Xin-Wei Shi [a,1], Li Liu [a,1], Jin-Ming Gao [a,b,c,*], An-Ling Zhang [a]

[a] Research Centre for Natural Medicinal Chemistry, College of Science, Northwest A&F University, Yangling, 712100 Shaanxi, China
[b] Shaanxi Engineering Center of Bioresource Chemistry & Sustainable Utilization, Northwest A&F University, Yangling, 712100 Shaanxi, China
[c] National Engineering Research Center for Phytochemistry in West China, Yangling, 712100 Shaanxi, China

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ABSTRACT

Two novel cyathane diterpenoids, designated scabronines K (**1**) and L (**2**), were isolated from the fruiting bodies of the basidiomycete *Sarcodon scabrosus* together with four known analogues, sarcodonins G (**3**), A (**5**), M (**6**), and scabronine H (**4**). Their structures were elucidated on the basis of extensive spectroscopic analysis including 2D-NMR (HMBC, HSQC, ROESY, ^{1}H,^{1}H-COSY) and MS experiments. The isolated compounds were evaluated for nerve growth factor (NGF)-mediated neurite outgrowth using rat pheochromocytoma (PC12) cells as a model system of neuronal differentiation. Among these compounds, only sarcodonins G and A (**3** and **5**) at 25 μM showed significant neurite outgrowth (neuritegenesis)-promoting activity in the presence of 20 ng/mL NGF after 24 h treatment. Their structure-neurite inducing activity relationship was also discussed.

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1. Introduction

Neurotrophic factors (neurotrophins) play a critical role in neuronal development, survival, and the functional maintenance of neurons [1–3]. They have been targeted as potential therapeutic drugs for the treatment of neurodegenerative disorders such as Alzheimer's disease (AD). Nerve growth factor (NGF) is the first and best characterized neurotrophic factor that is recognized as an important regulatory substance in the nervous system [4–6]. Thus, NGF is expected to have therapeutic efficacy for the treatment of AD. However, it cannot cross brain–blood barrier because of its high molecular polypeptide and is easily metabolized by peptidases under physiological conditions [7,8]. To address this issue, considerable efforts have been made to find small molecules that have neurotrophic properties and/or that are capable of enhancing the action of endogenous neurotrophic factors [7,9]. For instance, both hinokinin and 9-O-(11-hydroxyeudesman-4-yl)dihydrosesamin from the Japanese Cypress have been demonstrated to show potent neurite outgrowth-promoting activity in NGF-mediated PC12 cells [8].

The basidiomycete fungus *Sarcodon scabrosus* (Fr.) Karst, belonging to the family Hydnaceae, is a bitter mushroom. Previous chemical investigation of the genus *Sarcodon*, for instance *S. scabrosus* [10–15], *Sarcodon cyrneus* [16] and *Sarcodon glaucopus* [17] led to the isolation of a number of cyathane diterpenoid fungal metabolites. All these diterpenoids are unified by the presence of a characteristic 5/6/7 tricarbocyclic fused core structure. Many of these diterpene compounds showed interesting biological activities, such as antibiotic or antimicrobial [12,18–20], anti-MRSA (Methicillin-resistant *Staphylococcus aureus*) [21], agonistic toward the kappa-opioid receptor [22], anti-inflammatory [13,14], and anti-proliferative and apoptosis-inducible activities [23]. Importantly, however, both the erinacine and the scabronine natural products stimulated the synthesis of NGF in human nerve cells [11,24–29], which implicates their potential as therapeutic agents to treat neurodegenerative ailments such as Alzheimer's or Parkinson's disease.

In continuation of our ongoing search program for the isolation of bioactive natural products from higher fungi [30–35], we investigated the MeOH extract at 20 μg/mL of the fruiting bodies of the basidiomyceteous fungus *S. scabrosus* possessing pronounced

* Corresponding author. College of Science, Northwest A&F University, 3 Taicheng Road, Yangling, Shaanxi 712100, China.
E-mail address: jinminggao@nwsuaf.edu.cn (J.-M. Gao).
[1] These authors contributed equally to this work.

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neurite outgrowth-promoting activity in NGF-mediated PC12 cells, resulting in the isolation of two new cyathane diterpenoids, scabronine K (**1**) and scabronine L (**2**), as well as four previously reported analogues **3**–**6**. In this paper, herein we deal with the isolation and structure elucidation of compounds **1** and **2**, and NGF-potentiating activity of the isolated substances on PC12 cells, and discuss their structure-neurite inducing activity relationships.

2. Results and discussion

2.1. Isolation

The methanolic extract of the fruiting bodies of *S. scabrosus* (10 kg) was suspended in water and then partitioned with EtOAc. The EtOAc layer was repeatedly chromatographed over a series of silica gel, RP-18 silica gel, and Sephadex LH-20 to afford six metabolites (**1**–**6**) (Fig. 1). These included two new ones scabronines K (**1**) and L (**2**) and four known ones. Structures of the known isolated compounds scabronine H (**4**) [15], sarcodonin G (**3**) and sarcodonin A (**5**) [10], and sarcodonin M (**6**) (also known as scabronine B) [11,12], were established by comparing their physical and spectral data with those in the literature.

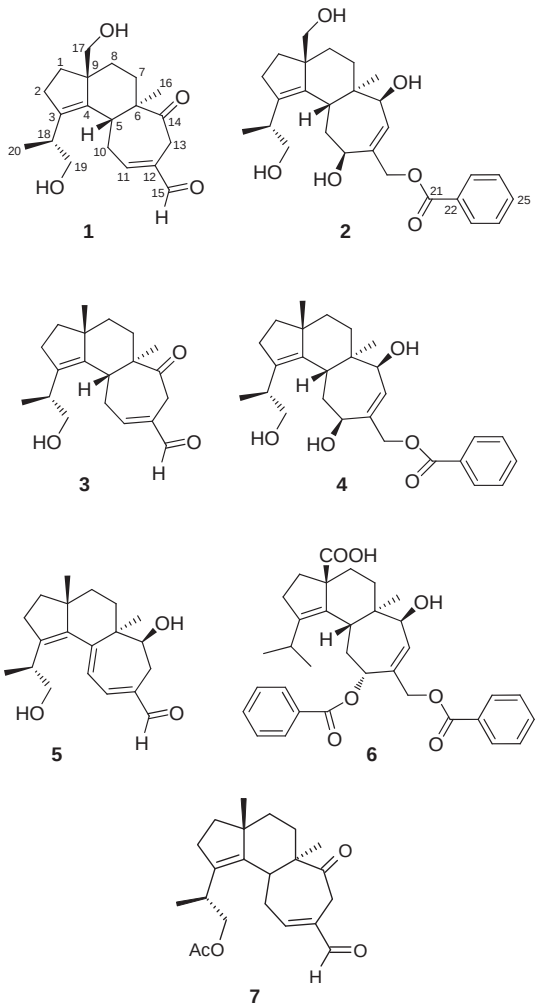

Fig. 1. Structures of scabronines K (**1**), J (**2**), and H (**4**), sarcodonins G (**3**), A (**5**), M (**6**), and 19-O-acetylsarcodonin G (**7**).

2.2. Structure elucidation

Scabronine K (**1**) was obtained as a yellow resin. Its molecular formula was established as $C_{20}H_{28}O_4$ by its positive HRESIMS (m/z 355.1880, $[M+Na]^+$; cacld for $C_{20}H_{28}O_4Na$, 355.1885), indicating 7 degrees of unsaturation. Its UV spectrum showed the existence of α,β-unsaturated system based on the absorption at 245 nm. The IR spectrum exhibited absorption bands at 3432, 1695 and 1684 cm^{-1} ascribable to OH, CO and α, β-unsaturated CO groups, respectively. The ^{13}C-NMR and DEPT spectra (Table 1) of **1** showed the presence of 20 carbon signals, which were recognized as two non-oxygenated sp^3 methines, six non-oxygenated sp^3 methylenes, two oxygenated sp^3 methylenes, two sp^3 quaternary C-atoms, one sp^2 methine, three sp^2 quaternary C-atoms, two sp^2 carbonyl C-atoms, and two methyl groups.

The ^{1}H-NMR spectrum of **1** recorded in CD_3OD (Table 1) showed signals for one secondary and one tertiary methyl groups at δ_H 0.99 (d, $J = 6.8$ Hz) and 1.01 (s, 3H), respectively. Moreover, the structural elements displayed in the ^{1}H- and ^{13}C-NMR spectra of **1** included an α,β-unsaturated aldehyde moiety [$\delta_{H/C}$ 6.87/155.6 (CH), 136.6 (C) and 9.36/194.3 (CH)], one tetrasubstituted double bond (δ_C 141.4 and 136.8), one oxygenated methylene ($\delta_{H/C}$ 3.45, 3.84/66.1), and one isolated carbonyl carbon (δ_C 213.0). The above spectral data allowed us to conclude that **1** may be a cyathane-type diterpenoid [10,15]. The ^{1}H, ^{1}H-COSY and HSQC spectra of **1** allowed the identification of four partial structures, **a**) $MeCHCH_2OH$, **b**) $-CH_2CH_2-$, **c**) $-CH_2CH_2-$, and **d**) $-CH-CH_2-CH-$ (Fig. 2). These partial structures were connected by the HMBC correlations (Fig. 2) of H-18/C-4 and C-2, H_2-17/C-1, C-9, and C-8, H_2-1/C-17, H-8/C-17, H-7/C-14, H_3-16/C-14, H_3-16/C-5, and H-13/C-14. These indicated that the C=C bond is located between C-3 and C-4 and that the isolated keton group is placed at C-14 (δ_C 213.0). The existence of the α,β-unsaturated aldehyde between C-11 and C-15 was implied by HMBC correlation of H-11 and H_2-13 with C-15. Moreover, the ^{1}H- and ^{13}C-NMR spectra of **1** are very similar to those of

Table 1
NMR data of scabronine K (**1**) in CD_3OD.[a]

No.	δ_H[b] (mult; J in Hz)	δ_C[b] (DEPT)	H,H-COSY[c] (selected)	ROESY[c] (selected)	HBMC[c] (selected)
1	1.44 (m); 2.04 (m)	34.2 (t)	H-2	H-2/H-17	C-2, 9, 17
2	2.35 (m)	30.2 (t)	H-1	H-1/H-17/ H-20	C-1, 3, 8, 9, 17
3		141.4 (s)			
4		136.8 (s)			
5	3.40 (m)	40.9 (d)	H-10	H_β-10/H-11	C-3, 6, 16
6		56.1 (s)			
7	1.69 (m); 2.01 (m)	33.8 (t)	H-8	H-8/H-16	C-8, 14, 16
8	1.56 (m); 1.73 (m)	32.7 (t)	H-7	H-7/H-17	C-7, 9, 17
9		56.3 (s)			
10	2.85 (m); 3.28 (m)	33.2 (t)	H-5/H-11	H-5/H-11/ H-16	C-5, 6, 11, 12
11	6.87 (m)	155.6 (d)	H-10	H-5/H-10/ H-15	C-5, 10, 15
12		136.6 (s)			
13	3.38 (m); 3.88 (m)	34.2 (t)			C-11, 12, 14, 15
14		213.0 (s)			
15	9.36 (s)	194.3 (d)		H-11	C-11, 12, 13
16	1.01 (s)	13.5 (q)		H-10/H-8	C-5, 6, 7, 14
17	3.45 (d, 11.2); 3.84 (d, 11.2)	66.1 (t)		H-1/H-8	C-1, 8, 9
18	3.13 (m)	36.7 (t)	H-19		C-2, 4, 19, 20
19	3.39 (m); 3.51 (m)	66.6 (t)	H-18/H-20		C-3, 18, 20
20	0.99 (d, 6.8)	16.4 (q)	H-18		C-3, 18, 19

[a] Assigned by ^{1}H NMR, ^{13}C NMR, DEPT, COSY, HSQC, ROESY, and HMBC experiments.
[b] Measured at 400/100 MHz.
[c] Measured at 600/150 MHz.

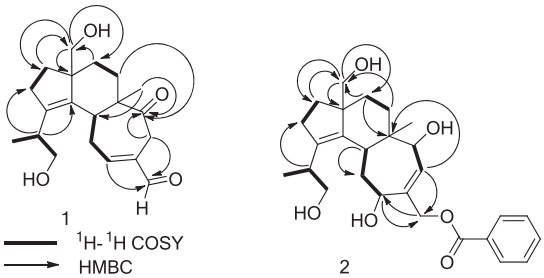

Fig. 2. Important ^{1}H-^{1}H COSY and HMBC correlations for scabronines K (**1**) and L (**2**).

sarcodonin G (**3**) [10]. The only difference was that a methyl group ($\delta_{H/C}$ 1.02/24.4) in **3** was replaced with an oxygenated methylene group in **1**. The relative configuration of **1** was determined by ROESY experiment (Fig. 3). Key NOE observed between H-5 and H-10β in **1** indicated that the H-5 is β-oriented. From these data, the structure of compound **1** was unambiguously established as 17,19-dihydroxy-14,15-dioxo-3,11-cyathadien, and named scabronine K.

Scabronine L (**2**) was obtained as an amorphous white powder. Its molecular formula was established as $C_{27}H_{36}O_6$ by its positive HRESIMS (*m/z* 479.2402, $[M+Na]^+$; cacld for $C_{27}H_{36}O_6Na$, 479.2409), indicating 10 degrees of unsaturation. The IR spectrum exhibited absorption bands at 3398, 1701, 1640, 1602 and 1584 cm^{-1} attributed to OH, ester CO groups and an aromatic ring, respectively. The ^{1}H-NMR signals of 5 aromatic protons at δ_H 7.48–8.03 and the ^{13}C-NMR signals of six carbons between δ_C 129.6 and 131.3, besides a carbonyl signal at δ_C 167.4, were consistent with a benzoate moiety.

The ^{13}C-NMR and DEPT spectra of **2** recorded in CD_3OD (Table 2) showed the presence of 27 carbon signals, in which two double bonds [δ_C 139.7 (s), 131.7 (s); 140.0 (s), 129.6 (d)], two non-oxygenated sp^3 methines (δ_C 41.7, 36.7), three oxygenated sp^3 methylenes (δ_C 63.9, 65.4, and 66.7), two oxygenated sp^3 methines (δ_C 75.1, 76.6), two sp^3 quaternary carbons (δ_C 43.5, 56.4), five non-oxygenated sp^3 CH_2 groups and two methyl groups were identified beside one benzoyloxy moiety. The ^{1}H- and ^{13}C-NMR spectral data of **2** are very similar to those of scabronine H (**4**) [15], except for the presence of a hydroxymethyl group [δ_H 3.40 (d, J = 11.2 Hz) and 3.85 (d, J = 11.2 Hz), and δ_C 65.4] instead of the methyl group at the C-9 position in **4**. The gross structure of **2** was further evidenced from ^{1}H, ^{1}HCOSY, HSQC, and HMBC experiments (Fig. 2). The HMBC spectrum of **2** displayed correlations of H-18/C-2; H-1/C-17; H-17/C-1, C-9, and C-8; H-8/C-6 and C-17; H-7/C-17; H-13/C-6 and C-15; H-11/C-15; and H-5/C-10. The relative configuration of **2** was assigned on the basis of ROESY correlations (Fig. 3). Correlations H-7α, H-8α, H-10α and H-13/Me-16 and H-13/H-14 of **2** indicated that these protons were situated on the same side. The correlation H-5/H-11 was not observed in the case of **2**, confirming the hydroxy group at C-11 to be β-oriented. Correlations of Me-16 with H-8α, H-10α and H-13, and of H-14 with H-13, displayed that **2** has the same configuration as that of **4**. Thus, the structure of **1** was determined to be 11,14,17,19-tetrahydroxy-3,12-cyathadiene-15-yl benzoate, and named scabronine L.

The absolute configurations for scabronines K and L (**1** and **2**) were determined as 5*R**, 6*R**, 9*S** and 18*R**, and 5*R**, 6*R**, 9*S**, 11*S**, 14*S** and 18*R**, respectively, based on the co-occurrence of structurally related known metabolites, sarcodonins A, G, and M, which

Table 2
NMR data of scabronine L (**2**) in CD_3OD.

No.	δ_H[a] (mult; *J* in Hz)	δ_C[a] (DEPT)	H,H-COSY[b] (selected)	ROESY[b] (selected)	HBMC[b] (selected)
1	1.38 (m); 2.00 (m)	33.9 (t)	H-2	H-2/H-17	C-2, 8, 9, 17
2	2.30 (m)	30.2 (t)	H-1	H-1/H-17/ H-20	C-1, 3
3		139.7 (s)			
4		131.7 (s)			
5	2.94 (br.d, 9.2)	41.7 (d)	H-10		
6		43.5 (s)			
7	1.22 (m); 1.92 (m)	33.3 (t)	H-8	H-8/H-16	C-17
8	1.52 (m); 1.62 (m)	33. 8 (t)	H-7	H-16/H-17	C-6, 9, 17
9		56.4 (s)			
10	2.24 (m); 2.48 (m)	34.5 (t)	H-5/H-11	H-16	C-5, 11, 12
11	6.14 (m)	75.1 (d)	H-10		
12		140.0 (s)			
13	3.85 (d, 6.5); 5.92 (d, 6.5)	129.6 (d)	H-14	H-14/H-15/ H-16	C-6, 14, 15
14	4.06 (d, 6.5)	76.6 (d)	H-13	H-13	
15	4.09 (d, 14.2); 4.20 (d, 14.2)	63.9 (t)		H-13	C-11, 12, 13
16	0.91 (s)	17.5 (q)		H-8/H-10/ H-13	C-5, 6, 7, 14
17	3.40 (d, 11.2); 3.85 (d, 11.2)	65.4 (t)		H-1/H-2/H-8	C-1, 8, 9
18	3.05 (m)	36.7 (d)	H-19/H-20		C-2, 3, 19, 20
19	3.30 (m); 3.47 (m)	66.7 (t)	H-18		C-3, 18, 20
20	0.97 (d, 6.8)	16.5 (q)	H-18		C-3, 18, 19
21		167.4 (s)			
22		129.6 (s)			
23,27	8.03 (t, 7.8)	130.1 (d)	H-24/H-26	H-15	C-21,24,25,26
24,26	7.48 (t, 7.8)	129.6 (d)	H-23/H-27/ H-25		C-22, 23, 27
25	7.60 (t, 7.8)	131.3 (d)	H-24/H-26		C-23, 27

[a] Measured at 400/100 MHz.
[b] Measured at 600/150 MHz.

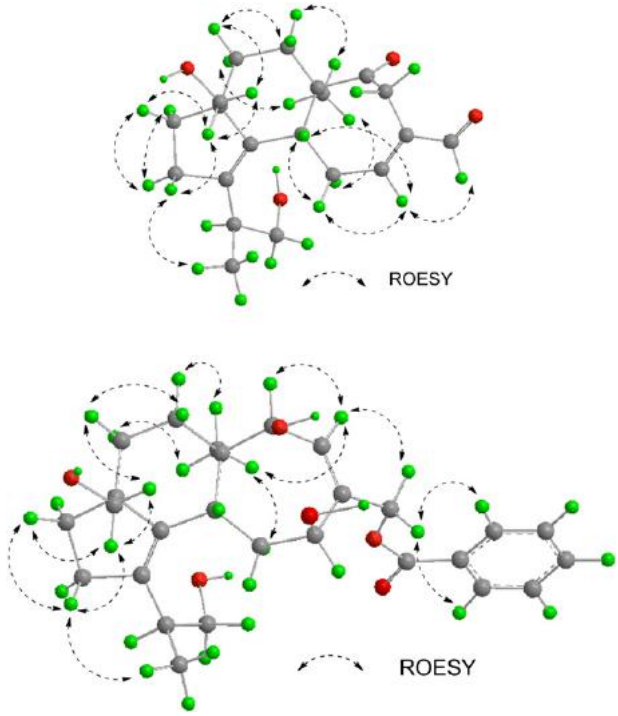

Fig. 3. Selected ROESY correlations observed for scabronine K (**1**) (the upper) and scabronine L (**2**) (the lower).

were reported previously from the same mushroom [10,11], and by comparison to their NMR data. By analogy with the cyathanes, it is assumed that the absolute stereochemistry is as shown in Fig. 1.

3. Neuritogenic activity

The neuritogenic activities of compounds **1**–**6** were examined using rat pheochromocytoma (PC12) cells as a model system of neuronal differentiation. The cells were incubated with the compounds **1**–**6** at a concentration of 25 μM in combination with NGF (20 ng/mL) as a positive control. Under these conditions, as shown in Fig. 4, only sarcodonins G (**3**) and A (**5**) exerted a pronounced increase in neurite-bearing cells compared to control cells (Fig. 4A), whereas the remaining metabolites including new ones (**1** and **2**) did not show such significant activity. The mean values expressed as percent of neurite-bearing cells in samples treated with 25 μM of sarcodonins G (**3**) and A (**5**) in combination of NGF (20 ng/mL) were 24.9% and 20.0%, respectively, as shown in Fig. 4B. The mean value of neurite-bearing cells in NGF treated control cells was 15.3%. Accordingly, co-incubation of cells with 20 ng/mL NGF and sarcodonins G (**3**) or A (**5**) resulted in a significant increase in neurite-bearing cells compared to cultures treated with NGF (20 ng/mL) alone. In contrast, sarcodonins G (**3**) or A (**5**) alone had no effect on neurite outgrowth in the absence of NGF (Fig. 4A). Additionally, in order to examine esterification of hydroxyl group at C-19 of **3** on the neurite inducing activity, we prepared a derivative of **3**, 19-O-acetylsarcodonin G (**7**), by acylation with dry Ac_2O in the presence of pyridine. Synthesized compound **7** was lacking in such activity. This indicates that the presence of the 19-OH is important for the potentiation of NGF-induced neurite outgrowth. The neurite outgrowth-promoting activity of sarcodonin G (**3**) toward PC12 cells was more significant than that of sarcodonin A (**5**). These results obtained indicated that among compounds **1**–**7**, two known compounds sarcodonins G (**3**) and A (**5**) can augment neurite outgrowth induced by NGF in PC12 cells. However, all the compounds **1**–**7** failed to induce neurite outgrowth in PC12 cells without NGF, The observations are consistent with the results reported for other scabronines, which were isolated previously from the same fungus [26,27].

It has been reported that other similar cyathane diterpenes, such as erinacines, induced NGF biosynthesis in rat astroglial cells [28] and that scabronines and cyrneines induce NGF expression in 1321N1 human astrocytoma cells [16,26,27] Taking into the structures of three kinds of cyathane diterpenes, cyrneines, erinacines, and scabronines account, it is very hard to summerize a structural feature that can justify the varying biological activities toward neurite outgrowth in PC12 cells [16,29].

Considering our observations concerning with compounds **1**–**7**, it appears that the presence of the α,β-unsaturated ($\Delta^{11(12)}$) aldehyde system in the seven-membered ring could be crucial for the activity on PC12 cells. Actually, scabronines and erinacines insert this structural feature and in our molecules **3** and **5** those that contain this same unit show a higher activity. Interestingly, it seems that the methyl group at C-9 is determinant, which is attributed to its low lipophilicity important for improvement of permeability to enhance penetration of the cellular membrane of PC12 cell, since the hydroxylation of this methyl group gave rise to a loss in the activity (e.g. **1** *vs* **3**). Furthermore, all the substitutions (hydroxylation at C-11 or a double bond at C-12 and C-13, and esterification at C-19) on cyathane carbon skeleton can also lead to a loss of the activity (e.g., **2**, **4**, **6** and **7** *vs* **3**). In other words, subtle differences in functional groups on cyathane structure can produce an impact on intereaction between PC12 cells and the small molecules. This is the first report of neurite outgrowth-promoting activity in PC12 cells shown by the two sarcodonins.

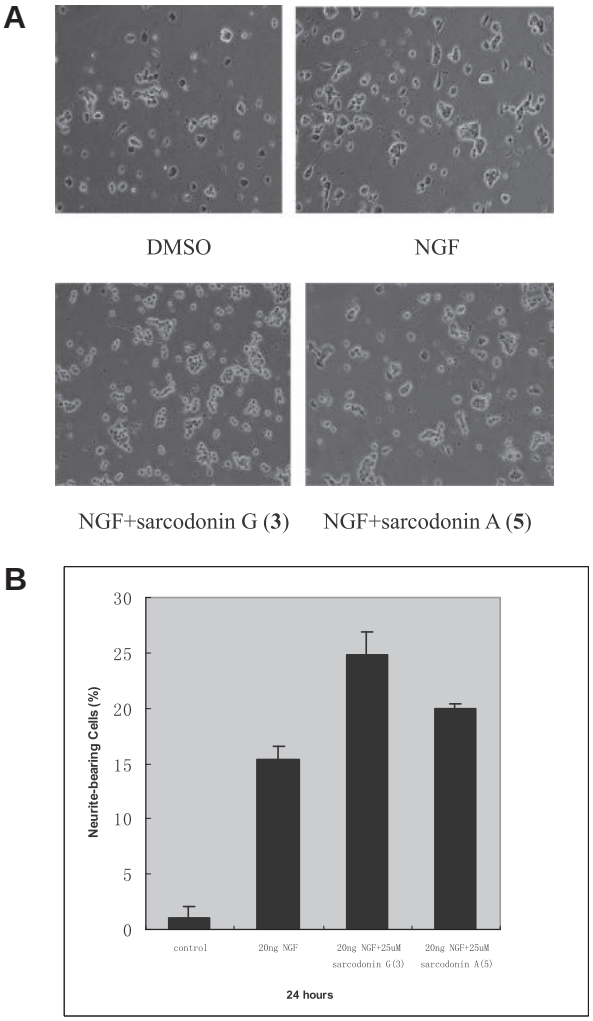

Fig. 4. Potentiation of NGF-induced neurite outgrowth in PC12 cells by sarcodonins G (**3**) and A (**5**) 24 h after treatment. (A) PC12 cells were treated in the presence of NGF (20 ng/mL), with sarcodonins G (25 μM) or A (25 μM), and NGF (20 ng/mL) alone then the morphological changes of the cells were observed. (B) PC12 cells were cultured in the presence of sarcodonins G (**3**) and A (**5**) (25 μM), or NGF (20 ng/mL) for 24 h. The bar graph represents percentage of neurite-positive cells after exposure to indicated concentrations of sarcodonins G and A, in the presence of NGF (20 ng/mL). All experiments were performed in triplicate. Values are means ± SEM of three wells. $^{*}P < 0.05$ versus control (without drug).

4. Experimental

4.1. General

Melting points were obtained on an X-4 micromelting point apparatus. Optical rotations were measured with a Horiba SEPA-300 polarimeter. UV spectra were obtained using a Shimadzu UV-2401A spectrophotometer. IR spectra were obtained by a Tenor 27 spectrophotometer using KBr pellets. 1D and 2D NMR spectra were recorded on Bruker AM-400, AVANCEIII-500 or AV-600 spectrometers with TMS as internal standard. Coupling constants were expressed in Hz, and chemical shifts (δ) were expressed in ppm with reference to the solvent signals. Mass spectra were performed on a VG Autospec-3000 spectrometer or an API Qstar pulsa TOF spectrometer.

Column chromatography (CC) was performed with silica-gel (200–300 mesh, Qingdao Marine Chemical Co. Ltd., Qingdao, China), Sephadex LH-20 (Pharmacia Fine Chemical Co., Uppsala, Sweden), Lichroprep RP-18 gel (40–63 μM, Merck, Darmstadt,

Germany). Fractions were monitored by TLC (GF_{254}, Qingdao Marine Chemical Co. Ltd., Qingdao, China), and spots were visualized by heating Si gel plates sprayed with 5% H_2SO_4 in ethanol.

4.2. Fungal material

The fresh fruiting bodies of *S. scabrosus* were collected at Ailao Mountain of Yunnan Province, P. R. China, in July 2002. The material was identified by Dr. Xiang-Hua Wang, Kunming Institute of Botany, the Chinese Academy of Sciences. Avoucher specimen was deposited in the Herbarium of the Kunming Institute of Botany, the Chinese Academy of Sciences.

4.3. Extraction and isolation

The fresh fruiting bodies of *S. scabrosus* (ca. 10 kg) were extracted three times with methanol (3 × 10 L, each 1d) at room temperature. After removal of the solvent under vacuum, the resulting methanol extract (300 g) was suspended in water and partitioned successively with ethyl acetate. The organic solution was dried over Na_2SO_4 and was concentrated in *vacuo*. The resulting EtOAc extract (200 g) was passed through silica gel column chromatography (CC) eluted with a gradient system of $CHCl_3$-acetone (9:1 → 1:9) to afford nine fractions (1–9) on the basis of TLC analysis.

Fraction 4 was chromatographed by a silica gel column, using $CHCl_3$–MeOH (98:2, 95:5, 90:10) as the elutent to obtain five subfractions (Fr.4.1–Fr.4.5). Fr.4.1 was purified by a Sephadex LH-20 column (MeOH) affording sarcodonin A (**5**) (250 mg). Fr.4.2 was purified by a Sephadex LH-20 column ($CHCl_3$/MeOH, 1:1) and then purified by preparative TLC ($CHCl_3$–acetone, 8:2) to yield scabronine K (**1**, 16 mg). Fr.4.3 was separated by a Sephadex LH-20 column (MeOH) followed by a RP-18 column (MeOH–H_2O, 8:2 → 9:1), and further purified by CC over silica gel eluting with $CHCl_3$–acetone (6:4) to furnish scabronine L (**2**, 13 mg).

Fraction 5 was separated by a Sephadex LH-20 column (MeOH) followed by CC on silica gel ($CHCl_3$-acetone, 95:5 → 80:20) to give five subfractions (Fr.5.1–Fr.5.5). Fr.5.1 was employed to a Sephadex LH-20 column ($CHCl_3$–MeOH, 1:1) followed recrystallization from MeOH to furnish sarcodonin G (**3**) (440 mg). Fr. 5.3 was applied to silica gel CC using $CHCl_3$–MeOH (100:3) to provide two subfractions (Fr.5.3.1 and Fr. 5.3.2). Fr. 5.3.1 was separated by Sephadex LH-20 (MeOH) followed by a RP-18 column using MeOH-H_2O (7:3–9:1), and further purified by CC over silica gel using $CHCl_3$-MeOH (9:1) to afford scabronine H (**4**, 18 mg). Fr. 5.3.2 was separated by CC over silica gel eluting with $CHCl_3$–MeOH (30:1) followed by Sephadex LH-20 ($CHCl_3$–MeOH, 1:1) to yield sarcodonin M (**6**, 8 mg).

4.3.1. Scabronine K (*1*)

Yellow resin ($CHCl_3$); [α] −36.4 (*c* 0.35, MeOH); UV (MeOH): λ_{max} 206, 245 nm; IR (KBr) ν_{max} 3431 (OH), 2920, 2851, 1695 (C=O), 1636 (C=O), 1444, 1040, 569 cm^{-1}; for ^1H and ^{13}C NMR data, see Table 1; positive HRESIMS *m/z* 355.1880 $[M+Na]^+$ (cacld for $C_{20}H_{28}O_4Na$, 355.1885).

4.3.2. Scabronine L (*2*)

Amorphous white powder (MeOH); mp 122–124 °C; [α] −5.4 (*c* 0.56, MeOH); UV ($CHCl_3$): λ_{max} 207.6, 233.7 nm; IR (KBr) ν_{max} 3398(OH), 2928, 2870, 1701(C=O), 1640, 1602, 1584, 1452, 1373, 1276, 1116, 1039, 1026, 937, 714, 687, 672 cm^{-1}; for ^1H and ^{13}C NMR data, see Table 2; positive HRESIMS *m/z* 479.2402 $[M+Na]^+$(cacld for $C_{27}H_{36}O_6Na$, 479.2409).

4.3.3. 19-O-Acetylsarcodonin G (*7*)

To a pyridine solution (0.4 mL) of sarcodonin G (**3**, 8.2 mg, 0.026 mmol) was added 0.2 ml of acetic anhydride at room temperature. The reaction mixture was stirred at 20 h. The residue after evaporation of the mixture was subjected to silica gel CC using petroleum ether-AcOEt (4:1) as the eluent to afford acetylated sarcodonin G (**7**, 6.3 mg). ESIMS (positive) *m/z*: 359.4 $[M+H]^+$.

4.4. Cell culture and evaluation of neurite outgrowth

Rat PC12 cells were sustained in Keighn's modified Ham's F-12 medium with 2 mM [SCAP] L-glutamine and 1500 mg/liter sodium bicarbonate supplemented with 15% horse serum, 2.5% (heat-inactivated) (fetal) bovine calf serum and 100 U/ml penicillin-streptomycin in a humidified atmosphere containing 5% CO_2 at 37 °C.

Before the differentiation experiments, cells were grown in this culture medium for 72 h and then singles cell PC12 suspensions were cultured in serum-free medium for the next 12–16 h. Each test compound and 20 ng/mL NGF (Wuan Hiteck Biological Pharma Co., Ltd, China) were dissolved in DMSO and PBS (pH 7.2), respectively. The final volume for all assays PBS or DMSO in each well, including controls was 0.1%. To rule out a potential influence of the solvent on the cellular parameters assessed, we conducted the corresponding control experiments.

Single cell PC12 suspensions were plated a density of 4×10^4 cells/well in serum-free culture medium and then treated with or without NGF at doses of 20 ng/mL and compounds and correct concentration, respectively. After 24 of incubation, the percentage of cells showing neurite outgrowth was determined by light microscopy. Cells with one or more neurites whose lengths were at least twice the diameter of the cell body were scored as positive. Neurite outgrowth was determined from at least three different regions of interest in three independent experiments.

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